

Domain Walls Become the Optimal Reversal of a Spin Chain: Free Optimal Control Beyond the Macrospin and an Energy-Barrier Floor on the Switching Cost

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Abstract

Optimal control of magnetization switching has been solved almost entirely for a single rotating moment, and its authors state in print that it remains open under what conditions nonuniform mechanisms such as domain-wall nucleation and propagation become optimal in terms of energy efficiency. We answer this for a ferromagnetic spin chain with nearest-neighbour exchange by minimizing the switching cost over the trajectory of every site, with no assumed reversal mode. Two results follow. First, a lower bound: inverting the equation of motion and applying the arithmetic-geometric mean inequality gives $\Phi \geq 4\alpha \Delta E / (\gamma\mu)$ for every pulse at every switching time, with ΔE the minimum-energy-path barrier; it equals the known infinite-time optimum for a single uniaxial moment. Second, above a crossover length and at long switching time the free optimum is a domain wall entering at an open end, strictly cheaper than uniform rotation. Over 61 chains ($J/K = 10$, N from 4 to 32, two damping values) 28 reverse more cheaply through a wall, by up to 51 percent, and every cost lies above its barrier floor. The saving at fixed switching time is not monotone in length: a longer wall needs more time to cross. One case was verified by grid refinement, local consistency, and open-loop forward dynamics. A comparison restricted to two fixed modes, uniform rotation against a constant-speed wall, reaches the opposite conclusion, because that wall is not the optimal one.

1 Introduction

Optimal control theory computes the applied-field pulse that reverses a magnetic moment at the least cost [1], and applied to two-dimensional van der Waals magnets it reaches picosecond, energy-efficient switching [2]. The formalism has been developed almost entirely for a single moment on the unit sphere. Its authors note the limitation: “A macrospin approximation is used in the present study, but this is expected to break down with increasing system size. Even if the initial and the final states are collinear, the transition between them may involve nonuniform rotation of magnetization such as nucleation and propagation of domain walls or excitation of spin waves. It remains to be seen under what conditions these and possibly other, yet unknown switching mechanisms become optimal in terms of energy efficiency” [3]. The only

step past the macrospin we know of is a one-dimensional study that found a standing spin wave in the optimal reversal above a critical length [4].

We minimize the switching cost of a spin chain over all trajectories, derive a barrier floor under the cost, and map where the optimum stops being uniform.

2 Model and cost

A chain of N unit moments $\vec{s}_i$, each of moment μ , has energy

$$E = -K \sum_i s_{i,z}^2 - J \sum_{\langle ij \rangle} \vec{s}_i \cdot \vec{s}_j, \quad (1)$$

with easy-axis anisotropy K per site, ferromagnetic exchange J , and open ends. The internal field at a site is $\vec{b}_i^{\text{int}} = -\mu^{-1} \partial E / \partial \vec{s}_i$. Given any trajectory, the Landau-Lifshitz-Gilbert equation is inverted at every site for the applied field that site needs,

$$\vec{b}_i = \frac{\alpha}{\gamma} \dot{\vec{s}}_i + \frac{1}{\gamma} \vec{s}_i \times \dot{\vec{s}}_i - \vec{b}_{i,\perp}^{\text{int}}, \quad (2)$$

and the switching cost is $\Phi = \int_0^T \sum_i |\vec{b}_i|^2 dt$, in T^2 s. It is not an energy; converting it to joules needs a circuit model, which we do not assume. Times are in units of $\tau_0 = \mu / (2\gamma K)$.

3 A barrier floor under the switching cost

Write $P_i = -\mu \vec{b}_i^{\text{int}} \cdot \dot{\vec{s}}_i$ for the power the internal field absorbs at site i , so that $\sum_i P_i = dE/dt$. Dotting Eq. (2) with $\dot{\vec{s}}_i$, and using $\dot{\vec{s}}_i \perp \vec{s}_i$, gives

$$\mu \vec{b}_i \cdot \dot{\vec{s}}_i = P_i + \mu \frac{\alpha}{\gamma} |\dot{\vec{s}}_i|^2. \quad (3)$$

Only the component of $\vec{b}_i$ along $\dot{\vec{s}}_i$ enters, so $|\vec{b}_i|^2 \geq (\vec{b}_i \cdot \dot{\vec{s}}_i / |\dot{\vec{s}}_i|)^2$. Where $P_i > 0$, the arithmetic-geometric mean inequality applied to the two terms gives $|\vec{b}_i|^2 \geq 4\alpha P_i / (\gamma\mu)$, and trivially $|\vec{b}_i|^2 \geq 0$ elsewhere. Summing over sites, $\sum_i |\vec{b}_i|^2 \geq (4\alpha/\gamma\mu) \max(dE/dt, 0)$, and integrating, the cost is at least $4\alpha/(\gamma\mu)$ times the total energy climbed. Every path from all up to all down passes through a configuration at least as high as the saddle of the minimum energy path, so it climbs at least that barrier ΔE , and

$$\Phi \geq \frac{4\alpha}{\gamma\mu} \Delta E \quad (4)$$

for every pulse and every switching time. For one uniaxial moment $\Delta E = K$ and the bound equals the infinite-time optimum $\Phi_\infty = 4\alpha K / (\gamma\mu)$ of Ref. [1], so it is tight there. Equality needs the field parallel to the velocity and the precession supplied by the internal field; uniform rotation meets both at long times, and whether a wall can is not guaranteed.

We compute ΔE with the climbing-image string method [5] in the geodesic form used for magnetic systems [6]. For $J/K = 10$ short chains reverse coherently ($\Delta E = NK$); above about eight sites a wall entering at an open end has the lower barrier, which saturates at $8.867 K$ against the continuum wall energy $2\sqrt{2JK} = 8.944 K$. The last row of Table 1 gives $\Delta E / (NK)$.

Table 1: Cheapest trajectory found over the uniform-rotation cost on the same grid, $J/K = 10$, $\alpha = 0.5$. Bold entries are below uniform rotation by more than one percent. The last two rows give the barrier floor at the longest switching time and the minimum energy path barrier relative to NK .

T/τ_0	$N = 4$	$N = 6$	$N = 8$	$N = 10$	$N = 12$	$N = 16$	$N = 20$	$N = 24$	$N = 32$
10	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
20	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
40	1.000	1.000	0.975	0.884	0.825	0.800	0.836	0.900	0.927
80	1.000	1.000	0.971	0.851	0.747	0.635	0.600	0.603	0.691
160	1.000	1.000	0.957	0.827	0.713	0.573	0.514	0.491	0.506
floor at 160	0.835	0.835	0.809	0.705	0.606	0.461	0.379	0.324	0.253
$\Delta E/(NK)$	1.000	1.000	0.968	0.844	0.725	0.552	0.443	0.369	0.277

Table 2: As Table 1 at $\alpha = 0.1$.

T/τ_0	$N = 8$	$N = 12$	$N = 16$	$N = 24$
20	1.000	1.000	1.000	1.000
60	1.000	1.000	1.000	1.000
150	0.977	0.878	0.934	1.000
300	0.972	0.773	0.709	0.787
floor at 300	0.918	0.687	0.524	0.350
$\Delta E/(NK)$	0.968	0.725	0.552	0.369

4 Free optimal control of the chain

The image-based direct minimization of Ref. [3] is generalized to N coupled sites: the trajectory is $Q + 2$ time slices of N unit vectors with clamped ends, the midpoint rule gives position and velocity on every interval and site, and Eq. (2) with the chain internal field gives the cost. The exact central-difference gradient is obtained in 36 cost evaluations independent of Q and N by graph coloring: an image touches two intervals in time and three sites in space, so images two apart in time and three apart in space are perturbed together. The cost is minimized by L-BFGS on normalized vectors from three starts, the single-site optimum on every site with a small perturbation, a tanh wall on the optimum’s azimuth, and the minimum energy path traversed on the optimum’s schedule. Uniform rotation is feasible and never pays exchange, so the uniform cost on the same grid is an exact upper bound on the optimum, and every cost we report is the cost of an explicit feasible trajectory: an upper bound on the true optimum even where the optimizer stopped early.

5 Results

Tables 1 and 2 give the cheapest trajectory found relative to uniform rotation. Three regimes appear.

Short chains. For $N \leq 6$ at $J/K = 10$ the barrier is NK and uniform rotation is optimal at every switching time: walls cannot lower the barrier, and no start finds a cheaper trajectory.

Short switching times. At $T \leq 20 \tau_0$ uniform rotation is optimal at every length even where the floor leaves large room. The wall has no time to cross the chain: its propagation cost grows with its speed.

Long chains at long times. Above the barrier crossover and at long switching times the optimum is nonuniform and strictly cheaper. At $\alpha = 0.5$ the saving reaches 51 percent ($N = 24$, $T = 160 \tau_0$), against a floor of 0.324 of the uniform cost. At $\alpha = 0.1$ the regime starts later,

near $150 \tau_0$, because the uniform optimum approaches its own infinite-time floor on a time scale that grows as the damping falls; at $300 \tau_0$ a sixteen-site chain saves 29 percent.

At fixed switching time the saving is not monotone in length. At $\alpha = 0.5$, $T = 40 \tau_0$ it is largest at $N = 16$ and shrinks for longer chains, and the optimal length moves up as T grows. A longer chain has a lower barrier per site but a longer distance for the wall to cross in the same time. Over all 61 chains every cost lies above its floor, the closest at 1.018 times it.

6 Verification of a sub-uniform trajectory

Because the result reverses a published conclusion, one case ($N = 16$, $J/K = 10$, $\alpha = 0.1$, $T = 150 \tau_0$) was checked three independent ways before being accepted. Its entry in Table 2 (0.934) comes from the baker's automatically chosen grid and fixed iteration budget; longer solves on finer grids lower it, which is why the values below differ. The minimum-energy-path start on a 300-slice grid gave 0.892 of the uniform cost. *Grid refinement*: interpolated to 600 slices and re-optimized, the ratio fell to 0.838 against the 600-slice uniform cost; a discretization artifact would have shrunk. *Local consistency*: integrating every interval from its left image under its own midpoint field with a fine Runge-Kutta step and comparing with its right image gives errors of the same order for the wall trajectory (maximum 4.1×10^{-3} , mean 7.1×10^{-4}) as for the uniform optimum (maximum 2.0×10^{-3} , mean 6.5×10^{-4}) on the same grid. *Open-loop dynamics*: applying the recovered per-site fields of the 600-slice trajectory to the chain equation of motion from the all-up state reverses every site (final $s_z = -1.000$ on all sixteen) and tracks the optimized trajectory to within 0.106. A second, independent start, the tanh wall alone, also lands below uniform rotation for this chain.

Three numerical traps were found on the way and are guarded in the code: the geodesic angle between nearly coincident images must be computed as $2 \arctan(|\vec{r} - \vec{l}|/|\vec{r} + \vec{l}|)$ rather than $\arccos(\vec{l} \cdot \vec{r})$, whose rounding error made the finite-difference gradient noise; the optimizer's relative stopping test must not be scaled by an expensive seed's own cost; and a wall seed must not compress the precession into each site's flip window, which creates near-antipodal jumps where the midpoint rule is singular.

7 Discussion and limits

The floor explains why the answer depends on length: the cost of any reversal is at least proportional to the barrier it must climb, and a wall entering at an open end climbs a barrier that stops growing with length while uniform rotation climbs NK . Whether a pulse can approach the floor is a question of time, because the wall must also be moved across the chain. The optimal reversal is therefore uniform for short chains or short times and wall-mediated for long chains at long times, with an optimal length that grows with the switching time.

The free search matters for this conclusion. Restricting the comparison to two fixed modes, a uniform rotation against a domain wall swept at constant speed, finds uniform rotation cheaper at every chain length at short switching time, and invites the reading that walls lower the energy barrier but not the field cost. That reading does not survive the free search: the constant-speed wall is not the optimal wall, the optimal wall is cheaper than uniform rotation at long switching time, and the barrier floor of Section 3 links the energy barrier and the field cost rather than opposing them.

The trajectories reported are the cheapest found, not certified global optima; most nonuniform solves stopped before their convergence criterion, so the true savings are at least those quoted. The study is one-dimensional, with nearest-neighbour exchange, a uniaxial anisotropy, and open ends; the floor holds in any dimension, but the crossover map is specific to this model. The dimensionless map depends only on J/K , α , T/τ_0 and N ; material parameters enter through τ_0 and the cost scale.

Data and code availability

The engine `spinoct` 0.10.0 (MIT, <https://pypi.org/project/spinoct/>, https://github.com/fsantibanezleal/CAOS_SpinOCT) provides the chain solver, the minimum energy path, and the floor. The crossover bake, the committed artifact `data/artifacts/lattice_ocp.json`, the table generator, and the web application are at https://github.com/fsantibanezleal/CAOS_RES_Espira; the live instance is at <https://espira.fasl-work.com>. Every number in Tables 1 and 2 is generated from the committed artifact.

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